

```

n = int(input())
arr = list(map(int, input().split()))

left = 0
max_length = 0
seen = set()

for right in range(n):
    while arr[right] in seen:
        seen.remove(arr[left])
        left += 1

    seen.add(arr[right])
    max_length = max(max_length, right - left + 1)

print(max_length)

```

```

5
1 2 3 1 2
3

```

```

n = int(input())
arr = list(map(int, input().split()))

current = arr[0]
maximum = arr[0]

for i in range(1, n):
    current = max(arr[i], current + arr[i])
    maximum = max(maximum, current)

print(maximum)

```

```

4
1 -2 3 1
4

```

```

n = int(input())
arr = list(map(int, input().split()))

left = 0
right = n - 1
left_max = 0
right_max = 0
water = 0

while left <= right:
    if arr[left] <= arr[right]:
        if arr[left] >= left_max:
            left_max = arr[left]
        else:
            water += left_max - arr[left]
            left += 1
    else:
        if arr[right] >= right_max:
            right_max = arr[right]
        else:
            water += right_max - arr[right]
            right -= 1

print(water)

```

```

6
0 1 0 2 1 0
1

```

```

n = int(input())
arr = list(map(int, input().split()))

current = arr[0]
maximum = arr[0]

for i in range(1, n):
    current = max(arr[i], current + arr[i])

```

```

    maximum = max(maximum, current)

print(maximum)

```

```

4
-2 1 -3 4
4

```

```

n = int(input())
arr = list(map(int, input().split()))

current_max = arr[0]
current_min = arr[0]
maximum = arr[0]

for i in range(1, n):
    x = arr[i]

    if x < 0:
        current_max, current_min = current_min, current_max

    current_max = max(x, current_max * x)
    current_min = min(x, current_min * x)

    maximum = max(maximum, current_max)

print(maximum)

```

```

5
2 3 -2 4 1
6

```

```

n = int(input())
arr = list(map(int, input().split()))

left = 0
maximum = 0
seen = set()

for right in range(n):
    while arr[right] in seen:
        seen.remove(arr[left])
        left += 1

    seen.add(arr[right])

    maximum = max(maximum, right - left + 1)

print(maximum)

```

```

8
1 2 3 1 5 4 2 3
5

```

```

n = int(input())
arr = list(map(int, input().split()))
target = int(input())

prefix_sum = 0
count = 0
seen = {0: 1}

for x in arr:
    prefix_sum += x

    if prefix_sum - target in seen:
        count += seen[prefix_sum - target]

    seen[prefix_sum] = seen.get(prefix_sum, 0) + 1

print(count)

```

```

5
1 2 3 2 1
5
2

```

```
n = int(input())
words = input().split()

groups = {}

for word in words:
    key = ''.join(sorted(word))

    if key not in groups:
        groups[key] = []

    groups[key].append(word)

for group in groups.values():
    print(*group)
```

```
6
eat ate sat hat fat bat dog
eat ate
sat
hat
fat
bat
dog
```

```
n = int(input())

appointments = []

for i in range(n):
    start, end = map(int, input().split())
    appointments.append([start, end])

appointments.sort()

merged = []

for start, end in appointments:
    if not merged or start > merged[-1][1]:
        merged.append([start, end])
    else:
        merged[-1][1] = max(merged[-1][1], end)

for start, end in merged:
    print(start, end)
```

```
4
1 3
2 5
3 6
4 8
1 8
```